Aryan Naik

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MEng Robotics and Artificial Intelligence, University College London (UCL) | Expected Graduation: 2029

Note: Applying for placement year entry or deferred 2027 cohort consideration. Available for industrial placement from June 2027 prior to final-year return.

EDUCATION

MEng Robotics and Artificial Intelligence | University College London (UCL) | 2024 - 2029

Relevant Coursework: Embedded Systems, Real-Time Computing, Autonomous Systems, Sensor Fusion, Digital Electronics, Control Theory

PROJECTS

alrs_007_capsule | C++ Firmware, Embedded Systems, Real-Time Control

- Achieved reliable parachute deployment sequencing by implementing a finite state machine in C++ firmware across a dual-MCU architecture, eliminating unhandled state transitions across all tested descent phases
- Achieved continuous in-flight sensor capture by integrating an IMU over I2C and configuring a structured data logging pipeline on a resource-constrained MCU, sustaining acquisition across the full flight envelope with no recorded data loss events during post-flight telemetry review
- Achieved deterministic sensor scheduling by partitioning processing responsibilities across two MCUs, reducing per-core interrupt load and enabling concurrent telemetry and control execution within a single embedded runtime

aegis_swarm | ROS 2, Distributed Autonomy, Simulation

- Achieved coordinated multi-agent behaviour across a distributed drone swarm by configuring FastDDS middleware for ROS 2 Jazzy, enabling reliable inter-node communication across all active swarm agents in simulation
- Achieved pre-deployment validation of swarm autonomy logic by building a digital twin environment in NVIDIA Isaac Sim, allowing fault scenarios and edge-case behaviours to be exercised in simulation before any hardware integration
- Achieved operator-in-the-loop teleoperation capability by integrating a Meta Quest 3 headset via OpenXR and Unity C#, enabling real-time manual override of autonomous swarm agents during simulation trials

SKILLS

Languages: C++, C, Python, C#

Frameworks and Middleware: ROS 2 Jazzy, FastDDS, OpenXR, Unity

Hardware and Embedded: Dual-MCU firmware architecture, I2C protocol, IMU integration, finite state machine design, real-time sensor data logging

Systems: NVIDIA Isaac Sim, digital twin simulation, distributed systems communication

Domains: Embedded computing, mission-critical state control, autonomous systems, aerospace sensor processing, open architecture system design

ADDITIONAL INFORMATION

Actively developing knowledge of FPGA-based processing pipelines and open architecture hardware standards relevant to rugged embedded computing and defence-grade sensor processing. Familiar with the design principles underlying high-reliability embedded systems through coursework in digital electronics and real-time computing.

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Hiring Team Abaco Systems

Dear Hiring Team,

I am writing to apply for the Abaco UK Graduate Scheme 2027. Abaco's work in rugged embedded computing and open architecture sensor processing for mission-critical defence platforms is precisely the engineering domain I am working towards. The combination of low-level hardware reliability and advanced signal processing that underpins your land, sea, and air programmes aligns directly with the technical direction of my degree and independent project work.

The project I consider most relevant to this role is my ALRS-007 capsule, a dual-MCU embedded system written in C++ for a high-altitude payload. I designed and implemented a finite state machine governing parachute descent control, integrated an IMU via I2C for real-time orientation logging, and built a telemetry pipeline to transmit sensor data throughout flight. The result was a functioning, flight-ready firmware architecture that operated reliably across the full mission profile. This project gave me direct experience of the constraints that define rugged embedded development: deterministic execution, sensor reliability, and fault-tolerant state management under conditions where failure is not recoverable.

Joining Abaco's graduate scheme would allow me to apply and deepen this embedded systems foundation within a professional defence engineering environment. I am particularly keen to develop my understanding of how open architecture standards are implemented at scale across long-lifecycle programmes, and how hardware and software co-design decisions are made under strict qualification requirements. I would bring strong C++ fundamentals, hands-on firmware experience, and a genuine interest in the engineering challenges specific to aerospace and defence. I am confident this scheme would be a formative step in my career.

Yours sincerely, Aryan Naik